MIL-R-81628B(AS)
1 April 1974
Superseding
MIL-R-81628A(AS)
of 16 July 1971

MILITARY SPECIFICATION

RADIO SET AN/ARC-143B

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide radio frequency reception and transmission on any of 7000 channels in 25 kilohertz (25 kHz) increments in the 225.000 through 399.975 MHz radio frequency (rf) band. Operationally, the equipment shall provide line-of-sight (LOS) and satellite (SAT) communications capability. More specific and detailed information is found in Section 3, REQUIREMENTS.

1.1.1 Line-of-Sight Mode - In the line-of-sight (LOS) mode, the equipment shall provide simplex two-way communications with the following modes:

a. Amplitude Modulation (AM) Normal Voice

b. Amplitude Modulation (AM) Secure Voice

c. Frequency Modulation (FM) Link 11, Parallel Tone, Phase-Shift Keyed Data

d. Frequency Modulation (FM) Secure Link 11 Data

e. Frequency Shift Keyed (FSK) Link 4, Serial Digital Data

Additionally, the equipment shall provide the capabilities for:

(1) Monitoring of the ultra high frequency (UHF) AM emergency channel by the incorporation of a separate fixed tuned guard receiver.

(2) Reception of UHF automatic direction finder (ADF) signals, on any selected channel.

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(3) Transmission of a signal from a reference signal generator of the Command Active Sonobuoy System (CASS) or Directional Command Active Sonobuoy System (DICASS) on a preselected channel.

1.1.2 Satellite Mode - In the satellite (SAT) mode the equipment shall provide half duplex operation. Receive frequencies shall automatically be offset from transmit frequencies. Three selectable and programmable transmit to receive frequency offsets shall be provided. The equipment shall also operate with an external modem via 70 MHz ports.

1.1.3 Built-In Test - Built-in test shall be available. Fault isolation to the functional module level shall be possible.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Paragraph
Radio Set	AN/ARC-143B	3.5.1, 3.5.2, 3.5.3
Receiver-Transmitter	RT-932B/ARC-143	3.5.1
Control, Radio Set	C-7791B/ARC-143	3.5.2
Mount, Electrical Equipment	MT-4064/ARC-143	3.5.3
Amplifier, Radio Frequency (Satellite Receiver Preamplifier)	AM-6699/A	3.5.4
SAT/LOS Select Panel	None	3.5.5

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in paragraph 6.8 as specified herein.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of contract form a part of this specification to the extent specified herein.

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SPECIFICATIONS

Military

MIL-B-5087	Bonding, Electrical, and Lightning Protection, for Aerospace Systems
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel, Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-M-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
NAVTACSTANS OPNAVINST C5711.91B	US Navy Book of Standards for Tactical Data Systems (This document is classified CONFIDENTIAL)

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STANDARDS

Military

MIL-STD-280	Definitions of Item Levels, Item Exchangeability Models and Related Terms
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-704	Electric Power, Aircraft, Characteristics, and Utilization of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-785	Reliability Program for System and Equipment Development and Production
MIL-STD-781	Reliability Tests, Exponential Distribution
MS-17322	Meter, Time Totalizing, Miniature, Digital

National Security Agency

CSEEB-13B	TSEC/KY-28 (V-2) Speech Security Equipment, Half Duplex Wide-Band, Airborne (U) (This document is classified CONFIDENTIAL.)

2.1.1 Availability of Documents - When requesting specifications, standards, drawings and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

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3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

b. Other parts and materials requirements shall conform to Specification MIL-E-5400.

c. Nonrepairable subassemblies shall be used as outlined in Specification MIL-E-5400. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (see 6.5).

d. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5. Conformal coatings, encapsulants, embedments, or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly. The provisions of Specification AR-5 shall govern microelectronic modular assembly interchangeability.

3.2.3 Modules - The electronic portions of the equipment shall be functionally modularized.

3.2.4 Solid-State Design - No electron tubes shall be utilized in the design of this equipment. Semiconductor transistors and diodes shall be in accordance with MIL-STD-454 Requirement 30.

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3.3 Design and Construction - The equipment shall conform with the applicable requirements of Specification MIL-E-5400 and MIL-STD-280 for design, construction and workmanship, except as otherwise specified herein.

The MIL-STD-454 requirement (limitation of no more than 3 wires per terminal) may be exceeded provided that:

a. The equipment performs satisfactorily during all testing and service use.

b. The wires are positioned on the terminal in such a manner that individual wires may be removed at any time without damage to the remaining wires or terminal post.

3.3.1 Total Weight - The total weight of the equipment, excluding SAT/LOS Select Panel, cables, and preamplifier assembly, shall be a minimum consistent with good design and shall not exceed 42 pounds.

3.3.2 Reliability- The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a re-order from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order. The reliability program shall be applied to both the development and production phases.

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 1200 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 20,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability In Mean-Time-Between-Failure (MTBF) - The equipment, including any Build-In-Test provisions, shall have 1200 hours specified mean (operating) time between failures, (θ_o), when tested and accepted as outlined under the requirements in 4.4.3, Reliability Assurance Test. The field MTBF, including externally induced failures, is expected to be in the 500 hour range.

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3.3.2.4 Time Totalizing Meter - The Receiver-Transmitter RT-932B/ARC-143 shall contain a time totalizing meter in accordance with Specification MIL-M-7793 and MS-17322.

3.3.3 Cabling and Connections -

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400. Functionally non-interchangeable, multi-pin connectors shall not be physically interchangeable.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements, provided the mating of the cable and its plugs may still be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipments.

3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400. The equipment shall be constructed to preclude the requirement for any "on aircraft" adjustments when replacing a Weapon Replaceable Assembly (WRA) by another WRA of the same type designation.

3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-I-6181, without the assistance of external suppression devices. The interference limits specified in MIL-I-6181 and the approved test plan shall apply to the generation of electromagnetic interference (EMI) by the transceiver and the susceptibility of the transceiver to an EMI environment, except as detailed in the contract.

3.3.7 Maintenance Provisions - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance. Built-in test features shall provide fault isolation to the functional assembly (see 3.6.2).

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3.3.7.1 Quantitative Maintainability Requirements - The quantitative maintainability requirements of the UHF Radio Set shall provide for the achievement of the following:

a. The equipment shall be designed so that faults shall be isolated to the radio set by means of built-in test features, computer diagnostic procedures or other means.

b. A Mean-Time-To-Repair (MTTR), at organizational level (WRA replacement), of 11 minutes, excluding access delay time and supply delay time.

c. A maximum time to repair ($M_{max\ ct}$) at organizational level of 17 minutes (95th percentile), excluding access delay time and supply delay time.

d. A Mean-Time-To-Repair (MTTR), at intermediate level of 30 minutes for the weapons replaceable assembly.

e. A maximum time to repair ($M_{max\ ct}$) at intermediate level of 33 minutes (95th percentile), for the weapons replaceable assembly.

f. A Mean-Time-To-Repair (MTTR), at intermediate level of 31 minutes for the repairable subassembly or functional module assembly.

g. A maximum time to repair ($M_{max\ ct}$) at intermediate level of 34 minutes (95th percentile), for the repairable subassembly or functional module assembly.

h. Probability of a correct fault isolation, utilizing the positions of the built-in test switch, shall be at least 80%.

3.3.7.2 Organizational and Intermediate Level, Maintenance Test and Checkout Requirements - The detail requirements shall be as specified in MIL-E-5400.

3.3.7.3 Test Points - Test points shall provide sufficient signal access to permit, as a minimum, field shop end-to-end tests and fault isolation to the functional module level.

3.3.7.4 Service and Access - The equipment shall be designed as follows:

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a. The WRA design shall consist of functional module assemblies wherever possible to facilitate fault isolation and repair by replacement at the field level.

b. The mechanical design of WRA's, functional module assemblies and cabling shall preclude incorrect assembly, connection or installation.

c. Repairable subassemblies shall be contained within the functional module assemblies wherever possible.

3.3.8 Nomenclature and Nameplates - Nomenclature and serial number assignment, nameplate approval, and identification marking shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used for making laboratory bench tests:

Temperature	Room Ambient (25°C ±10°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90% Relative Humidity
Input Power Voltage	
AC Power	115/200 ±1 VAC, 3 Phase, 400 Hz
AC Power	5 VAC (Panel Lighting)
DC Power	18 volts DC (GO Indicator Lamp)

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as modified herein. Service conditions define the environment beyond the limits specified for standard conditions.

3.3.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400 except as modified herein. (Mounted both with and without vibration isolators.)

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3.3.11 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed one (1) minute under standard conditions and two (2) minutes under service conditions.

3.3.12 Input Electrical Power -

3.3.12.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704, and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704.

The power required shall not exceed the specified amounts:

RT-932B: AC Power (three phase) 115/200 V, 400 Hz, Category "B", 800 VA.

C-7791B: AC Power (single phase) 5 V, 10 VA (Panel Lighting)
DC Power 18 V, 1 watt (Panel GO Lamp)

3.3.12.2 Voltage Transients, Frequency Transients and Power Interruptions -

a. During normal operation, as per MIL-STD-704, of the primary electronic system, utilization equipment shall provide 100 percent performance within the following limits of MIL-STD-704:

Limits 2 and 3 of Figure 3
Limits 2 and 3 of Figure 5

b. The utilization equipment shall withstand the following voltage and frequency transients and power interruptions due to faults external to the power supply, improper adjustment of the equipment itself, or associated equipment. The equipment shall withstand any overload without permanent damage to the equipment and shall return to normal operation once the overload has cleared by operation of the Control Box ON-OFF mode selector switch.

(1) AC voltage transients shall be within limits 1 and 4 of Figure 3 of MIL-STD-704.

3.3.12.3 Protective Devices - Electronic current overload protective devices shall be used. Fuses or circuit breakers shall not be used.

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3.3.12.4 Short Circuit Protection - The equipment shall be designed so that a short circuit external to the equipment shall not cause permanent damage to the equipment.

3.3.13 Cooling -

3.3.13.1 Thermal Conditions - The Radio Set and associated items as defined in 1.2 shall operate satisfactorily under the following thermal conditions:

a. MIL-E-5400, Class 1, for the R/T unit with attached airborne cooling assembly.

b. MIL-E-5400, Class 1, for the Satellite Receive Preamplifier and Electrical Equipment Mount.

c. MIL-E-5400, Class 1, for the Radio Set Control and the SAT/LOS Select Panel.

3.3.13.2 Thermal Design - The equipment shall conform to its specified performance and reliability characteristics under a MIL-E-5400, Class 1 environment as defined in 3.3.13.1. The R/T unit shall be designed to use a cooling assembly (blower) for movement of the cooling air. R/T unit blower cooling and operating performance characteristic curves shall be provided for the cooling blower assembly.

3.3.13.2.1 R/T Cooling in Aircraft - A removable cooling assembly attached to and within the maximum outline dimensions of the R/T unit shall normally be provided to supply adequate cooling air flow under MIL-E-5400, Class 1 environmental conditions, as defined in 3.3.13.1.a.

3.3.14 Pressurization - The equipment shall not require or utilize pressurization.

3.3.15 Antenna VSWR Protection - Electronic circuits shall be used to automatically protect the transmitter circuits from damage whenever the load VSWR exceeds 3:1. Similarly, the receiver circuitry shall be undamaged by any value of antenna VSWR.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under standard conditions. When reduced performance under service conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified. Otherwise, standard performance shall be required.

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3.4.1 Operation - The transceiver covered by this specification is an ultra-high frequency (UHF) airborne radio transceiver capable of continuous duty on high performance military aircraft under extreme environmental and operational conditions. It shall provide simplex radio transmission and reception on any one of 7000 channels in 25 kHz increments in the 225.000 to 399.975 megahertz (MHz) radio frequency (rf) band. The transceiver shall include a separate guard receiver for reception of emergency voice signals on the standard emergency frequency of 243 MHz with provision for changing to other frequencies in the rf band between 238.00 and 248.00 MHz.

The transceiver shall be capable of operation in the following modes and shall be compatible with NAVTACSTANS Volume 1 and 2 (OPNAVINST C5711.91B), unless otherwise specified herein:

a. Amplitude Modulation (AM) - Normal Voice

b. Amplitude Modulation - Wideband Encrypted Voice

c. Amplitude Modulation - Automatic Direction Finding.

d. Frequency Modulation (FM) - Parallel Tone Phase-Shift Keyed Data (Link 11).

e. Frequency - Shift Keying (FSK) - Serial Digital Data (Link 4)

f. Built-In Test (BIT)

g. Amplitude Modulation - Sonobuoy command (transmit only)

h. External Modem (Interface with an external modem shall be possible via 70 MHz ports.)

The UHF radio transceiver shall contain an automatically tuned, multichannel receiver-transmitter and a separate guard receiver. All modes, frequencies and channels shall be remotely controlled. The equipment shall be capable in the line-of-sight mode of providing both reception and transmission of AM Voice (normal and secure), FM (Link 11) data (normal and secure), and FSK (Link 4) data. The equipment shall be compatible with Automatic Direction Finding Group AN/ARA-48, AN/ARA-50, and AN/ARA-25. Additionally, the equipment shall be compatible with an Audio Command Signal Generator for transmission of sonobuoy command and control messages.

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In the satellite mode the radio set shall be capable of operation with the satellite terminal equipment including operation with a DPSK modem (per ELEX-M-103 or equivalent) via 70 MHz ports. Also three transmit-to-receive frequency offsets, wire programmable from 40.1 to 60.0 MHz in 100 kHz increments, shall be provided in the satellite band. The local oscillator injection shall be provided at a receiver-transmitter port for external use. An external port shall be provided to permit receiver-transmitter operation from an external 1 MHz frequency reference.

3.4.2 Frequency Characteristics -

3.4.2.1 Frequency Range - The main receiver and transmitter of the radio set shall provide 7000 communication channels in the radio frequency range commencing with 225.000 MHz and extending to 399.975 MHz in 25 kHz increments. There shall be no more than 42 degraded channels. A degraded channel is defined in 3.5.1.5.1.3.

3.4.2.2 Frequency Accuracy - The injection frequencies in the equipment for both the main receiver and transmitter shall be derived from either an internal or external 1 MHz frequency reference standard. Using the internal standard, the overall frequency accuracy shall be at least 1 part in 10^5 when operated from -55°C to -40°C, and at least 5 parts in 10^6 when operated from -40°C to +70°C. The overall frequency accuracy shall equal the frequency accuracy of the external standard when operated with this standard.

3.4.2.3 Frequency Stability -

3.4.2.3.1 Long-Term - The long-term frequency stability of the complete equipment at operating temperature shall be better than 1 part in 10^6 per month when operating with the internal frequency standard.

3.4.2.3.2 Short-Term - When operated with an external 1.0 MHz oscillator having a stability better than 1 part in 10^8 over 10 milliseconds, the radio set shall have a short-term frequency stability better than 1 part in 10^8 over an averaging period of 10 milliseconds. Under service conditions, the short-term stability shall be better than 8 parts in 10^8 over an averaging period of 10 milliseconds when the radio set is subjected to vibration without vibration isolators.

3.4.3 Terminations -

3.4.3.1 Receiver Input - The rf input to the receiver shall be 50 ohms nominal with a VSWR not to exceed 2.5 to 1.

3.4.3.2 Transmitter Load - The transmitter load impedance shall be 50 ohms nominal with a VSWR not to exceed 3.0 to 1.

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3.4.3.3 Transmitter VSWR Protection - Protective circuitry shall be incorporated such that when the VSWR exceeds 3:1, the transmitter circuitry shall be protected from damage. Normal operation shall be resumed, when the fault is removed, by re-keying the transmitter.

3.4.4 Receiver Outputs -

3.4.4.1 Normal AM - The normal AM audio load impedance shall be 600 ohms ±10%, resistive, balanced. This output shall be floating with respect to chassis ground. External grounding any one of the output loads shall have no adverse effects on receiver performance.

3.4.4.1.1 Audio Muting - The audio output of the receiver shall be muted during the period of channeling or frequency selection.

3.4.4.2 Wideband AM - The wideband AM output shall be terminated in a single-ended 20,000 ohms (±10%) resistive load.

3.4.4.3 FM Data - The FM data load impedance shall be 600 ohms ±10% resistive. This output shall be floating with respect to ground.

3.4.4.4 FSK Data - The FSK data output shall be terminated in a single-ended 2200 ohms ±10% resistive load.

3.4.4.5 Guard Receiver KY-28 Output - The guard receiver output shall be terminated in a single-ended 600 ohm ±10% resistive load.

3.4.4.6 70 MHz IF Modem Port - The 70 MHz IF port from the radio set to the external modem shall be designed to operate into a 50 ohm termination, and provide a minimum bandwidth of 500 kHz at a response 1 dB below the center frequency response.

3.4.4.7 First L.O. Injection - The first L.O. injection (70 MHz below channel frequency) shall be available at a port on the receiver-transmitter. The output level shall nominally be 0 dBm into a 50-ohm termination and shall not decrease below -6 dBm under service conditions. When this line is terminated at the R/T port, the antenna conducted emanations at LO frequency shall be kept to a minimum and not degrade the system performance. Double shielded cabling should be used in external wiring for this application.

3.4.5 Transmitter Modulation Inputs -

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3.4.5.1 Voice Input - The voice input circuit of the transmitter shall present an unbalanced, ungrounded, resistive load of nominally 100 ohms to the signal source. When connected to a dummy microphone of 100 ohms resistance, a carbon microphone energizing current of 55 ±10 milliamperes shall be supplied.

3.4.5.2 FM Data Input - The data input circuit of the transmitter shall present a balanced, ungrounded, resistive load of nominally 600 ohms ±10% to the signal source.

3.4.5.3 FSK Data Input - The FSK input circuit of the transmitter shall present a grounded resistive load of 2200 ohms ±10% to the signal source.

3.4.5.4 Sono Input - The sono input circuit of the transmitter shall present an unbalanced 100 ohms ±10% resistive load to the signal source.

3.4.5.5 70 MHz Modem Modulation Input to the Receiver-Transmitter - The 70 MHz modem modulation input to the receiver-transmitter shall present a nominal impedance of 50 ohms. The receiver-transmitter shall provide a minimum 3 dB bandwidth of 500 kHz.

3.4.6 Transmitter Keying - A ground $(^{+0.2V}_{-0.0V})$ on the transmit keyline, capable of sinking 5 ma, will place the radio set in the transmit mode. An open circuit on the keyline will place radio set in the receive mode. The keyline circuits within the radio set shall withstand transients on the keyline in the receive mode which do not exceed +5 volts peak or -0.1 volt peak.

3.4.6.1 Sono Transmit - Transmission shall be controlled by the keyline.

3.4.7 Ancillary Equipment Interfaces -

3.4.7.1 Control, Receiver-Transmitter Interface - Mode, frequency, and squelch disable control information shall be transmitted from the remote control to the receiver-transmitter as serial digital information on a single twisted pair bus. In addition, initiation of self-test shall be transmitted serially on the same twisted pair as mode and frequency control. Timing information is generated within the control and supplied to the R/T on a separate line.

3.4.7.1.1 Control Box Channel Changing Time - The time from the instant any channel is selected on the control box to the time that the R/T is operative for either transmission or reception shall not exceed 2 seconds.

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3.4.7.2 Data Terminal Set (Link 11)/ Radio Set Interface - Interface between the radio set and the Link 11 data terminal set shall be as follows:

3.4.7.2.1 Radio Set Automatic Key - The data terminal set will provide the transmit key voltages to the radio set on a single unbalanced to ground keyline per Paragraph 3.4.6.

3.4.7.2.2 Data Audio Lines - Data audio lines between the radio set and the data terminal set shall have the following characteristics:

a. A pair of receiver data audio lines, 600 ohms, transformer coupled.

b. A pair of transmitter data audio lines, 600 ohms, transformer coupled.

3.4.7.2.3 Data Audio Characteristics - The transmitted or received data audio will be composite preamble and message tones with a nominal rms level of 0 dBm and a peak-to-rms ratio of 14.7 dB.

3.4.7.2.4 Transmit/Receive Switching - Upon application of the transmit key control signal, to the equipment input terminals, the equipment shall switch to the transmit function, and shall be stabilized within 1 dB of rated power within 7 milliseconds after application of the transmit key signal. Upon removal of the transmit key control signal, the equipment shall switch to the receive function and provide full receiver specification performance within 7 milliseconds.

3.4.7.3 Digital Data Communications Set (Link 4)/Radio Set Interface - Interface between the radio set and the digital data communications set (Link 4) shall be as follows:

3.4.7.3.1 Receive to Transmit Switching Requirements (Controlled Station Mode) - The equipment shall meet the following requirements for switching from receive to transmit operation.

3.4.7.3.2 Receive to Turnaround Operation - The equipment shall remain in receive condition when either an open circuit or $+5\ ^{+0.25}_{-0.50}$V is applied to the "Turnaround" input circuit. Application of a "Turnaround" pulse of $^{+0.4}_{-0}$ volts shall cause the equipment to be switched to the prepared-to-transmit condition. The equipment shall accommodate input signal rise and fall times which are no slower than 2 microseconds as measured between the 10% and 90% amplitude points.

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3.4.7.3.3 Turnaround Keying Time - Turnaround operation, from receive to prepared-to-transmit condition, shall not exceed 2 milliseconds after application of the "Turnaround" pulse to the input circuit. Turnaround operation from transmit to maximum sensitivity receive condition shall not exceed 10 milliseconds.

3.4.7.3.4 Leakage Power - Leakage rf power from the transmitter shall not exceed -85 dBm at the operating frequency when in "Turnaround" condition.

3.4.7.3.5 Carrier-On Operation - The transmitter will be keyed by a key signal as defined in paragraph 3.4.6.

3.4.7.3.6 Carrier-On Keying Time - Maximum specified rf power shall be applied to the output load within 160 microseconds after application of each "carrier-on" pulse. The rf power in the output load shall decrease to not more than -85 dBm within 80 microseconds after cessation of the "carrier-on" pulse.

3.4.7.3.7 Turnaround and Carrier-On Pulse Termination -

a. Turnaround Input - During the application of the turnaround pulse, (0 to +0.2 V), the maximum current through the keying source shall not exceed 5 milliamperes. Receive conditions shall be caused by an open circuit or by +5 (+0, -1.6) VDC.

b. Carrier-On Input - During the application of the carrier-on signal, (0 to +0.2 V), the current through the keying source shall not exceed 5 milliamperes. Receive conditions shall be caused by an open circuit or 28 VDC.

3.4.7.3.8 Data Characteristics - The transmit or receive data shall be nominally +5 volts for +20 kHz deviation, and -5 volts for -20 kHz deviation.

3.4.7.4 ADF Interface -

3.4.7.4.1 ADF Operation - The equipment shall operate in conjunction with Radio Direction Finding Group AN/ARA-48 or AN/ARA-50, and AN/ARA-25 to provide bearing information. When used for this service, the receiver will receive radio frequency signals from the direction finding antenna, amplify and demodulate them and deliver the low-frequency modulation components to the direction finding group. The receive audio output characteristics required are described in paragraph 3.5.1.5.9.2.

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3.4.7.4.2 ADF Time Constant Line - Upon selection of the ADF mode of operation by the control, a signal shall be provided to the receiver-transmitter from the ADF equipment on a single wire. The signal shall be a ground capable of sinking 10 milliamperes. In all other modes, this signal shall be an open circuit (greater than 10,000 ohms resistive).

3.4.7.5 External Modem/Radio Set Interface (SAT) - Interface between the radio set and the external modem shall be as follows:

3.4.7.5.1 70 MHz Modem Ports - In the satellite mode a 70 MHz IF output from the radio set shall be provided. The output level shall be -88 dBm +35 dB into 50 ohms. A 70 MHz modulation input shall be provided to the radio set. Input levels from the modem shall be 0 dBm +1 dB into 50 ohms, and the radio set shall provide a minimum bandwidth of 500 kHz at a response 3 dB below the center frequency response.

3.4.7.5.2 Transmitter Keying - The transmit key to the radio set shall be on a single unbalanced to ground keyline. A ground ($^{+0.2}_{-0.0}$V) on the transmit keyline, capable of sinking 5 mA, shall place the radio set in the transmit mode. An open circuit on the keyline shall place the radio set in the receive mode. The keyline circuits within the radio set shall withstand transients on the keyline in the receive mode which do not exceed +5 volts peak or minus 0.1 volt peak.

3.4.7.5.3 Transmit/Receive Switching - Upon application of the transmit key control signal, the equipment shall switch to the transmit function and shall be stabilized within 1 dB of rated power output within 20 milliseconds maximum.

3.4.7.6 1 MHz Frequency Standard Interface - The radio set external 1 MHz frequency standard input shall present a nominal impedance of 50 ohms. The input level from the external oscillator shall be nominally 0 dBm measured across 50 ohms.

3.4.7.7 Local Oscillator Interface - The first local oscillator shall be available at an external port, in the satellite mode, when the radio set is used with a predetection combiner requiring local oscillator input. The local oscillator output, when available, shall be nominally 0 dBm into a 50 ohm load.

3.4.7.8 Satellite System Control Wires - The following signals shall be available at the receiver-transmitter output signal connector on separate wires.

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A logic "1" is 3.4 V minimum capable of supplying 50 microamperes.

A logic "0" is ground (0.35 V maximum) capable of sinking 5 milliamperes.

a. LOS Satellite Frequency Offset - Two unbalanced to ground wires, binary encoded, shall command the R/T into the LOS or the SAT mode. Three of the possible four codes transmitted on the two wires shall command the R/T into the satellite mode, and in addition when the R/T is in satellite mode receive, select one of three receive satellite frequency offsets. Logic to select frequency offsets shall be wire programmable in the field. Selection of SAT 1 or SAT 2 or SAT 3 mode of operation shall be accomplished by activating either SAT CONTROL LINE 1 or SAT CONTROL LINE 2 or both lines together to override all LOS mode commands either from Radio Set Control C-7791B/ARC-143, or from the computer referenced in paragraph 3.5.2.5. The desired satellite mode shall be selected by activating the SAT control lines in accordance with the following:

Receiver-Transmitter Mode	CONTROL LINE STATUS SAT Control Line 1	SAT Control Line 2
LOS	Logic "1"	Logic "1"
SAT 1	Logic "1"	Logic "0"
SAT 2	Logic "0"	Logic "1"
SAT 3	Logic "0"	Logic "0"

b. SAT/LOS Antenna Switch Control - The antenna switch control line shall be provided to control switching between the line of sight (LOS) antenna and the SAT antenna. The radio set shall accommodate a maximum switching time of 10 ms.

The SAT/LOS antenna control shall be a logic level "0" R/Toutput in the SAT 1, SAT 2 or SAT 3 modes to enable the satellite antenna signal path. It shall be a logic level "1"R/T output in all LOS modes to enable the LOS antenna signal path.

c. L.O. Output Enable - In the satellite mode, the receiver-transmitter shall be commanded to activate rf signal output from the local oscillator port by applying a logic level "1" on the L.O. output enable

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control line. A logic level "0" command on this control line shall inhibit L.O. signal output from the receiver-transmitter port.

d. High Power Amplifier (HPA) Control - In the satellite mode, the HPA control shall be a logic level "0" R/T output during transmission to enable the HPA signal path. In the receive condition, it shall be a logic level "1" R/T output to disable the HPA signal path.

e. Pre-Amp Receiver-Switch-Bias - The preamplifier receiver-switch-bias in the transmit condition shall be +115 volts from a 27 K ohm R/T source impedance to inhibit the receive signal path. In the receive condition, a bias current greater than 100 milliamperes, but limited to less than 220 milliamperes, shall be returned to a -6 volt R/T sink to enable the receiver signal path.

f. Pre-Amp Transmitter-Switch-Bias - The preamplifier transmitter-switch-bias in the receive condition shall be +12 volts from a 5600 ohm R/T source impedance to inhibit the transmitter signal path. In the transmit condition a bias current greater than 100 milliamperes, but limited to less than 220 milliamperes shall be returned to a -6 volt R/T sink to enable the transmitter signal path.

3.4.7.9 Headset/Microphone - The equipment shall operate in conjunction with a Headset H-173 with attached microphone M-96A/A and associated Electrical Cord Assembly, with in-line microphone amplifier and PTT switch CX-7889/AIC-22(V), or equivalent. The interface connections shall be as shown in Figure 1 herein. Wiring from the Headset/Microphone connector shall be paralleled with the radio interface to the aircraft intercommunication set AN/AIC-22 or equivalent.

3.5 Detail Requirements -

3.5.1 Receiver-Transmitter, Radio - The receiver-transmitter (WRA) shall meet the following requirements.

3.5.1.1 Function - The receiver-transmitter, shall provide the specific capabilities listed in 3.5.1.5 through 3.5.1.7 and provide the necessary control and excitation to satisfy the particular applicable requirements of 3.4.

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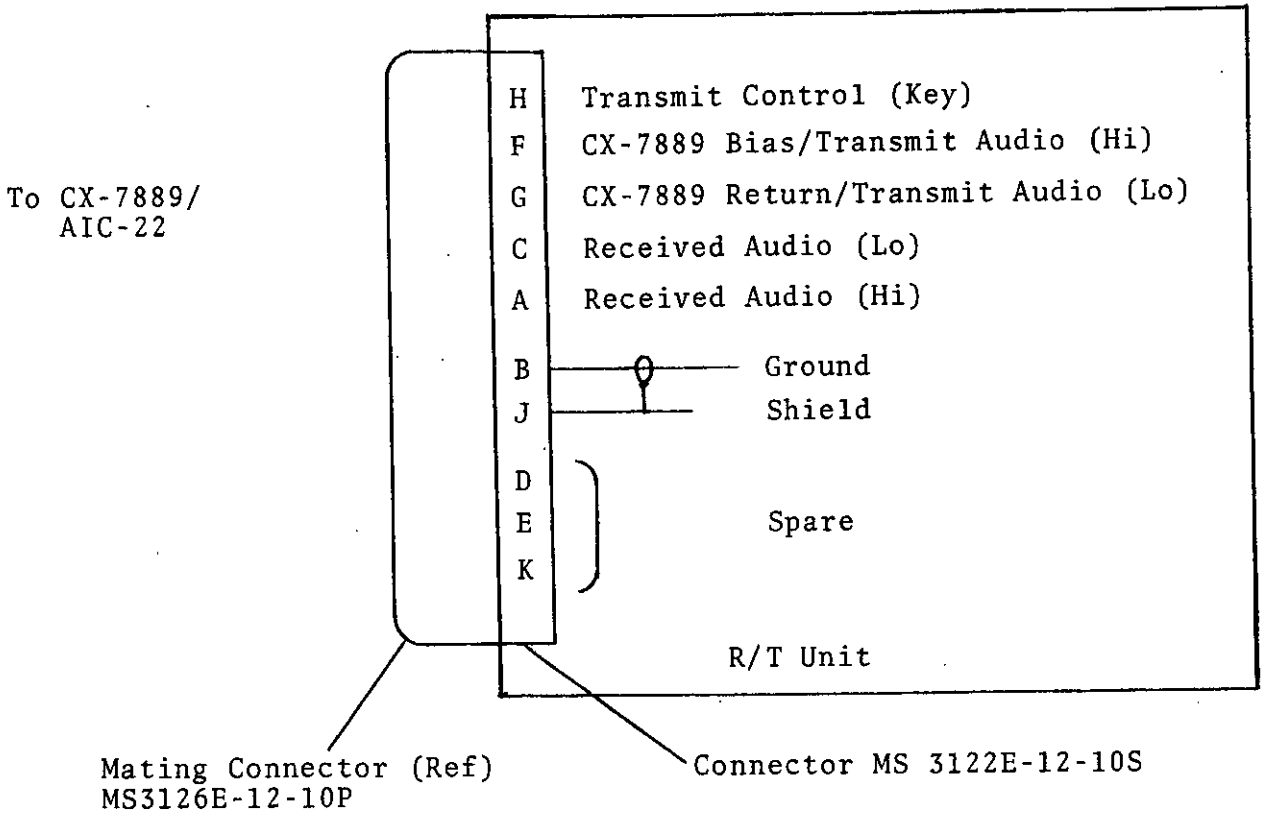


Figure 1. Interconnection Diagram of R/T Headset/Microphone Connector

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3.5.1.2 Form Factor - The receiver-transmitter shall be enclosed in a case. Overall dimensions including blower assembly and front panel extension shall not exceed 6.1 inches x 7.8 inches x 28.0 inches. (See Figure 2.)

3.5.1.2.1 Rear Guard - The receiver-transmitter shall have a rear handle to facilitate handling and protect the blower motor.

3.5.1.3 Weight - The weight of receiver-transmitter shall not exceed 36 pounds, including Blower and Assembly.

3.5.1.4 Contents - The receiver-transmitter shall contain the following functional module assemblies, interconnected by a frame and harness assembly.

a. Power Supply - The power supply shall generate and regulate major low voltage sources used in the equipment. It also shall contain EMI filtering and provision for self-testing.

b. Frequency Synthesizer/Control - The frequency synthesizer/control shall generate all injection frequencies used in both the receiver and transmit frequency conversions, shall accept serial bit stream data and clock inputs for purposes of controlling channel, mode, self-test and radio set functions, and shall also accept modulation inputs for FM and FSK. It shall also provide frequency offset for satellite operations. A 70 MHz input port shall be provided to accept modulation input from the 70 MHz modem. The frequency synthesizer shall lock up within 20 milliseconds maximum.

c. Receiver - The receiver shall contain the necessary circuitry for double conversion superheterodyning of the AM, FM, FSK antenna signals in the prescribed frequency range. An output port shall be provided to feed the 70 MHz received signal to an external modem. In addition, the receiver shall provide for the reception of the Guard frequency. The receiver shall also contain self-test circuitry. The receiver shall operate at the offset frequency in the satellite mode.

d. Transmitter - The transmitter shall be comprised of an exciter and a broadband power amplifier. The exciter shall generate the AM signal from baseband inputs and provide rf drive to the power amplifier.

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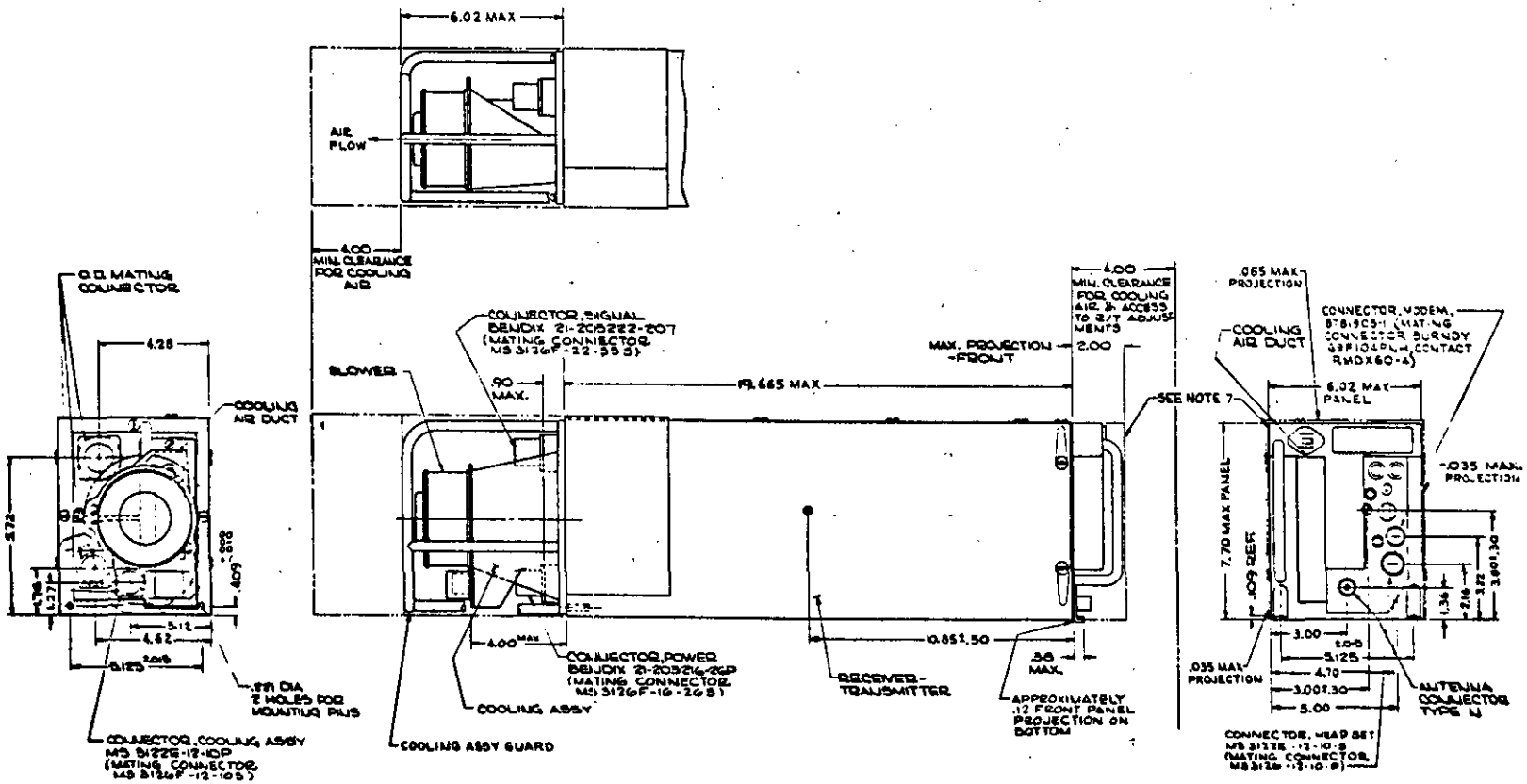


Figure 2. Outline and Mounting for Receiver-Transmitter, Radio RT-932B/ARC-143

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The power amplifier shall amplify the exciter signal within the 225.000 to 399.975 MHz frequency range to its final power level.

e. Blower - Provides the required equipment cooling. (See Paragraph 3.3.13.)

3.5.1.5 Main Receiver Characteristics - Under standard conditions, the main receiver noise figure shall not exceed 12 dB.

3.5.1.5.1 Receiver Sensitivity -

3.5.1.5.1.1 FM Sensitivity - The sensitivity of the receiver under standard conditions and over the frequency band of interest shall be such that a 20 dB signal plus noise/noise ratio is produced with an input signal of 2.0 microvolts (open circuit), frequency modulated to +6 kHz peak deviation by a 1 kHz sinusoidal tone. A signal level of not more than 4.0 microvolts (open circuit) shall produce a 20 dB signal plus noise/noise ratio under service conditions. When a measurement for this specification is performed, a post-detection filter shall be used which has a passband of 300 to 3,000 Hz. This filter shall be connected externally.

3.5.1.5.1.2 FSK Sensitivity - The sensitivity of the receiver under standard conditions and over the frequency band of interest shall be such that a 20 dB signal plus noise/noise ratio is produced with an input signal of 2.0 microvolts (open circuit) frequency modulated to +20 kHz peak deviation by a 1 kHz sinusoidal tone. A signal level of not more than 4.0 microvolts (open circuit) shall produce a 20 dB signal plus noise/noise ratio under service conditions. When a measurement for this specification is performed, a post-detection filter shall be used which has a cut-off frequency of 10 kHz. This filter shall be connected externally.

3.5.1.5.1.3 Main (AM) Sensitivity - With the squelch disabled, the sensitivity of the receiver under standard conditions and over the specified frequency range shall be such that an input signal of not more than 3.0 microvolts (open circuit), modulated 30 percent at 1 kHz applied at the antenna terminals of the UHF radio set, shall produce a minimum signal plus noise to noise ratio of 10 dB and a minimum audio output of 50 milliwatts across a resistive load of 600 ohms. A signal level of not more than 6.0 microvolts (open circuit), shall produce a minimum signal plus noise to noise ratio of 10 dB under service conditions.

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3.5.1.5.1.3.1 Receiver Interference - Minimum interference shall be produced in the guard receiver due to oscillators (and their harmonics) required for operation of the main receiver except when the main receiver is set 14-15 MHz or 70 MHz above the guard frequency. Interference in the main receiver shall be minimized except at harmonics of the guard receiver local oscillator. Channels on which the main or guard receiver sensitivity or automatic gain control characteristics or both are degraded because of spurious signals that are caused by harmonics or combinations of harmonics of oscillators in the radio set should be placed in one of the classifications given as follows:

Class	Sensitivity in Microvolts
1	greater than 3.0 but less than 5.0
2	greater than 5.0 but less than 10.0
3	greater than 16.0 but less than 30.0
4	30 or more

Channels which may be degraded shall be identified and be common to all receivers of the same nomenclature.

3.5.1.5.2 Selectivity -

3.5.1.5.2.1 AM Wideband/FM Selectivity - The overall selectivity of the receiver shall be within the limits set forth below:

Attenuation Below Response at Nominal Center Frequency	Total Bandwidth
6 dB	70 kHz Min.
60 dB	190 kHz Max.

Peak-to-valley ratio shall not be more than 2.5 dB over the 70 kHz IF bandwidth.

3.5.1.5.2.2 AM Narrowband Selectivity - The overall selectivity of the receiver shall be within the limits set forth below:

Attenuation Below Response at Nominal Center Frequency	Total Bandwidth
3 dB	38 kHz Min.
60 dB	90 kHz Max.

Passband ripple shall not exceed 2.5 dB over 38 kHz bandwidth.

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3.5.1.5.3 Cross-Modulation LOS - A 3.0 microvolts (30 percent modulated at 1000 Hz) on-channel signal shall produce a signal plus noise to noise ratio of at least 6 dB in the presence of off channel signals which are modulated 30 percent at 400 Hz and which are at levels and frequency separations from the desired signal in accordance with the following:

Interfering Signal Level dB above 3.0 Microvolts	Frequency Separation from Desired Signal
70	plus or minus 3.0 MHz
55	plus or minus 2.0 MHz
50	plus or minus 1.0 MHz
40	plus or minus 0.5 MHz
30	less than plus or minus 0.5 MHz, but outside IF passband

3.5.1.5.4 Image and Spurious Rejection - Main receiver image rejection shall be at least 70 dB down from center frequency response, spurious responses shall be at least 70 dB below.

3.5.1.5.5 Intermediate Frequency Rejection - The main receiver shall provide an attenuation of at least 80 dB relative to the response of the desired signal to a signal on the intermediate frequencies.

3.5.1.5.6 Local Oscillator Radiation - Antenna conducted emanations shall be in accordance with MIL-I-6181 except that the first mixer injection frequency and/or the harmonics of the injection frequency shall not exceed 50 microvolts.

3.5.1.5.7 Regeneration - The interstage and overall regeneration of the receiver shall be so minimized that no evidence of instability, unnecessary oscillation, or excessive regeneration shall occur at maximum gain.

3.5.1.5.8 Receiver Amplitude Variation (AGC) - The AGC characteristics shall be such that the maximum variation of the audio output is within 3 dB for rf input signals (30 percent modulated at 1000 Hz) from 3 to 100,000 microvolts and within 5 dB from 3 to 500,000 microvolts. Under service conditions, 6 dB shall be allowed from 6 to 100,000 microvolts and 8 dB from 6 to 500,000 microvolts. A maximum attack time of 100 milliseconds and a minimum release time of 20 milliseconds apply.

3.5.1.5.8.1 FSK - With an input signal at ±20 kHz peak deviation at a 1000 Hz rate, the output level shall not vary more than 10% for input signal level variations from limiter saturation to 20 dB above this level. The limiter shall be saturated with signal input to the receiver of -80 dBm.

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3.5.1.5.8.2 FM (Link 11) Data - Receiver audio output shall not vary more than +2 dB for rf inputs from 3.0 microvolts to 500 millivolts (open circuit) deviated 20 kHz peak with a 1930 +5% Hz tone.

3.5.1.5.9 Audio Output Characteristics -

3.5.1.5.9.1 AM Narrowband (Main Audio Output)

3.5.1.5.9.1.1 Level - The receiver shall be capable of delivering an output of at least 250 milliwatts in the AM mode with an rf input of 1000 microvolts modulated 30% by a 1000 Hz tone.

3.5.1.5.9.1.2 Response - The receiver audio output shall be within +1, -3 dB of a 1000 Hz reference for modulation signals between 300 Hz and 4 kHz.

3.5.1.5.9.1.3 Distortion - Total harmonic distortion at the receiver normal audio output shall not exceed 5% with an AM input signal of 1000 microvolts modulated 30% with an 1 kHz tone.

3.5.1.5.9.1.4 Internal Noise Level - The noise output of the receiver under squelched conditions shall be at least 40 dB below the standard condition audio output (250 milliwatts) produced by a 1000 microvolt signal modulated 30 percent at 1000 Hz per second. The signal plus noise to noise ratio of the receiver shall be at least 30 dB when a 1000 microvolt (open circuit) signal modulated 30 percent at 1000 Hz is applied and the audio output (signal plus noise) is adjusted to 250 milliwatts.

3.5.1.5.9.2 AM Wideband -

3.5.1.5.9.2.1 Level - The receiver audio output shall be 0.25 volt minimum with an rf input of 1000 microvolts modulated 30 percent at 1000 Hz.

3.5.1.5.9.2.2 Response - The receiver audio output shall be within +2, -3 dB of a 1000 Hz reference for modulating signals between 300 Hz and 25 kHz. Under service conditions, the audio response shall not vary from the response at 1 kHz by more than +3, -4 dB over the range of 300 Hz to 25 kHz.

3.5.1.5.9.2.3 Phase Shift - The wideband audio output shall not be more than 20 degrees out of phase with the negative portion of the rf signal modulation envelope when the rf signal is modulated 30% at 100 Hz. This output shall not pass through the carrier-to-noise squelch and shall be operative at all times.

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3.5.1.5.9.3 FM Data (Link 11) -

3.5.1.5.9.3.1 Level - For a 1000 microvolt signal deviated 30 kHz peak at a 1.93 kHz rate, the FM data output shall be 3 V RMS ±1 dB. Under service conditions, the FM output shall be 3 V RMS, ±3/2 dB in response to the same input signal at ambient temperatures below 20°C.

3.5.1.5.9.3.2 Stability Response - The instantaneous frequency error at the receiver audio output shall not exceed 0.5 Hz or 2.38 degrees phase shift in any 22 millisecond period over the frequency range of 900 Hz through 2.5 kHz.

3.5.1.5.9.3.3 Intermodulation Distortion - Any intermodulation product resulting from two-tone test at 10 kHz peak deviation per tone shall be at least 35 dB below the output level resulting from either tone. The two tones shall consist of 935 Hz and 1045 Hz.

3.5.1.5.9.3.4 Internal Noise Level - The receiver output hum and noise shall be at least 45 dB below the nominal audio output (3 V RMS) when measured in a 55 Hz bandwidth with an input signal of 1000 microvolts (open circuit), deviated ±30 kHz at a 1.93 kHz rate.

3.5.1.5.9.3.5 Envelope Delay - Overall envelope delay for any tone in the range of 300 to 3100 Hz shall not exceed 500 microseconds.

3.5.1.5.9.3.6 Response - From 300 Hz to 3000 Hz, the audio response to an input signal deviated 6 kHz shall not depart more than 2 dB from a minus 6 dB octave straight line (on a dB amplitude vs. log frequency plot) passing through a reference at 1000 Hz. Test equipment linearity error shall not be included.

3.5.1.5.9.4 FSK Data (Link 4) -

3.5.1.5.9.4.1 Level - The receiver shall be capable of delivering an output of 10 volts, plus 2, minus 0 volts peak-to-peak into the data output load with an input signal having an rf level sufficient for limiter saturation and frequency modulated sinusoidally at ±20 kHz deviation with a 5000 Hz rate. The data DC reference level shift shall not exceed ±2.5 volts under all conditions of equipment frequency tolerances and environmental changes and under all conditions of control station frequency stability and tolerance, the limits of which are ±2.5 kHz.

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3.5.1.5.9.4.2 Response - The audio response shall not vary from the response at 1 kHz by more than +2 dB over the range of 300 Hz to 5 kHz with a sine wave frequency modulation and +20 kHz deviation. Test equipment frequency linearity error shall not be included.

3.5.1.5.9.4.3 Receiver Output Linearity - The data output voltage shall be linear within +10% of the maximum peak-to-peak value data output voltage for signal deviations from zero up to +20 kHz.

3.5.1.5.10 Squelch,AM - A squelch threshold adjustment shall be located on the front panel of the receiver-transmitter. With the control in the maximum CCW position, the receiver shall be unsquelched. With the control in the maximum CW position, no more than 20 microvolts shall be required to unsquelch the receiver. Under standard conditions the adjustment range of this control shall be such that the squelch threshold can be set to correspond to an output signal plus noise to noise ratio between the limits of at least 6 dB and 15 dB when the rf signal input is modulated 30 percent at 1000 Hz. The squelch circuit shall have positive action so that the receiver audio is either completely off or completely on at all times. Unless specified otherwise, the squelch shall be adjusted to open at a 6 dB signal plus noise to noise ratio. Means shall be provided in the control system for operator on-off control of the squelch. (See 3.5.2.7.)

3.5.1.6 Guard Receiver Characteristics -

3.5.1.6.1 Sensitivity - With the squelch disabled, the sensitivity of the receiver under standard conditions and over the specified frequency range shall be such than an rf input signal of not more than 4.0 microvolts (open circuit) modulated 30 percent at 1 kHz, applied at the antenna terminals of the UHF radio set, shall produce a minimum signal plus noise to noise ratio of 10 dB. A signal level of not more than 8.0 microvolts (open circuit) shall produce a minimum signal plus noise to noise ratio of 10 dB under service conditions.

3.5.1.6.2 Selectivity - The overall selectivity of the receiver shall be within the limits set forth below:

Attenuation Below Response at Nominal Center Frequency	Total Bandwidth
6 dB	55 kHz Min.
60 dB	120 kHz Max.

Passband ripple shall not exceed 2.0 dB over the 3 dB bandwidth.

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3.5.1.6.3 Cross-Modulation - A 4.0 microvolt (30 percent modulated at 1000 Hz) on-channel signal shall produce a signal plus noise to noise ratio of at least 6 dB in the presence of off channel signals which are modulated 30 percent at 400 Hz and which are at levels and frequency separations from the desired signal in accordance with the following:

Interfering Signal Level dB above 4.0 Microvolts	Frequency Separation from Desired Signal
60	plus or minus 3.0 MHz
55	plus or minus 2.0 MHz
45	plus or minus 1.0 MHz
40	plus or minus 0.5 MHz
30	0.5 MHz, but outside IF passband

3.5.1.6.4 Image and Spurious Rejection - Image rejection shall be at least 60 dB, and spurious response shall be at least 60 dB below the response at center frequency.

3.5.1.6.5 Receiver Amplitude Variation - The AGC characteristics shall be such that the maximum variation of the audio output is 3 dB for 30 percent modulated at 1000 Hz rf input signals from 4 to 100,000 microvolts and a 5 dB variation from 4 to 500,000 microvolts. Under service conditions, 6 dB shall be allowed from 8 to 100,000 microvolts, and 8 dB from 8 to 500,000 microvolts. The maximum attack time shall be 100 milliseconds and the minimum release time shall be 20 milliseconds.

3.5.1.6.6 Audio Output Characteristics -

3.5.1.6.6.1 Level - The auxiliary guard output (KY-28 auxiliary output) voltage shall be at least 0.25 volt when a 1000 microvolt signal modulated 30% at 1000 Hz is applied to the rf input terminal. The receiver shall be capable of delivering a minimum output of 80 milliwatts to the main receiver audio output terminals when in the T/R+G mode.

3.5.1.6.6.2 Response - The receiver audio output shall be within +1, -3 dB of a 1000 Hz reference for modulation signals between 300 Hz and 4 kHz.

3.5.1.6.6.3 Distortion - Total harmonic distortion of the guard signal at the main or guard receiver audio output shall not exceed 5 percent for a 1000 microvolt signal modulated 30 percent with a 1 kHz tone.

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3.5.1.6.7 Squelch, AM - A squelch threshold adjustment shall be located on the front panel of the receiver-transmitter. With the control in the maximum CCW position, the receiver shall be unsquelched. With the control in the maximum CW position no more than 20 microvolts shall be required to unsquelch the receiver. Under standard conditions the adjustment range of each control shall be such that the squelch threshold can be set to correspond to an output signal plus noise to noise ratio between the limits of at least 6 dB and 15 dB when the RF signal input is modulated 30 percent at 1000 Hz. The squelch circuit shall have positive action so that the receiver audio is either completely off or completely on at all times. Unless specified otherwise, the squelch shall be adjusted to open at a 6 dB signal plus noise to noise ratio. Means shall be provided in the control system for operator on-off control of the squelch. (See 3.5.2.7.)

3.5.1.7 Transmitter Characteristics - The transmitter shall employ broadband tuning throughout the frequency range of 225.000 to 399.975 MHz. No electromechanical tuning shall be utilized.

3.5.1.7.1 RF Power Output - The transmitter shall be capable of producing into a 50 ohm resistive load 30 watts of AM carrier power, $^{+2}_{-1}$ dB, and 100 watts of FM carrier power $^{+2}_{-1}$ dB. Under service conditions the transmitter shall be capable of producing 30 watts of AM carrier power $^{+2}_{-3}$ dB and 100 watts of FM carrier power $^{+2}_{-3}$ dB into a 50 ohm resistive load. Automatic level control shall be used to limit the variation in output power to within 2 dB peak-to-peak across the frequency band.

3.5.1.7.2 Duty Cycle - The R/T shall be capable of continuous transmission under standard conditions. In addition, the unit shall be capable of continuous transmission for 30 minutes at sea level in accordance with the Class 1 environmental requirements of MIL-E-5400.

3.5.1.7.3 Transmitter VSWR Load - The transmitter shall provide specified output with normal power amplifier loading when terminated in a 50-ohm load with a VSWR not to exceed 1.3:1. The transmitter power output shall not be reduced by more than 3 dB additional for a VSWR of 2.5:1.

3.5.1.7.4 Frequency Accuracy - The final carrier frequencies shall be electronically derived from either the internal frequency standard or from the external frequency standard. Using the internal standard, the overall frequency accuracy shall be better than 1 part in 10^5 from -55°C to -40°C and at least 5 parts in 10^6 when operated from -40°C to 70°C. The overall frequency accuracy shall equal the frequency accuracy of the external standard when operated with this frequency standard.

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3.5.1.7.5 Spurious and Harmonic Radiation - The power of any spurious output 100 kHz removed from the carrier frequency but less than or equal to ±10% of the frequency of the carrier, shall be at least 55 dB below the carrier level. At any frequency greater than ±10% of the carrier frequency, any spurious output shall be at least 60 dB below the carrier level, except for the second harmonic of the carrier frequency which shall be at least 50 dB below the carrier.

3.5.1.7.6 Modulation Characteristics (Input) -

3.5.1.7.6.1 AM Modulation -

3.5.1.7.6.1.1 Input Level - The equipment shall be capable of modulating the rf output carrier not less than 90% when the modulating input is between -20 and +10 dBm. The average modulation percentage shall be calculated from the formula:

$$\frac{E_{max} - E_{min}}{E_{max} + E_{min}} \times 100\%$$

3.5.1.7.6.1.2 Response -

a. AM Normal Voice (Narrowband) - The audio response shall not vary from the response at 1 kHz by more than +1, -3 dB over the range of 300 Hz to 4 kHz.

b. AM Secure Voice (Wideband) - The audio bandwidth shall be 300 Hz to 25 kHz, with a maximum variation of +1, -3 dB relative to a 1 kHz reference over the range of 300 Hz to 25 kHz.

3.5.1.7.6.1.3 Distortion - The total harmonic distortion of the detected rf output envelope shall not exceed 10% for any tone in the 300 Hz to 25 kHz audio range, when the equipment is adjusted for 80% modulation at 1 kHz and a carrier level of 30 watts.

3.5.1.7.6.1.4 Input Impedance - The AM Voice Normal/Wide input circuit of the transmitter shall be a two wire transformer coupled resistive load of 100 ohms ±10% ungrounded. A single wire control line having a ground level $(^{+0.2}_{-0.0})$ volts shall provide the AM Wide modulation frequency response. An open circuit on this control line shall provide the AM Normal modulation frequency response.

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3.5.1.7.6.1.5 Noise Level - The audio voltage detected from an unmodulated 30 watt carrier shall be at least 40 dB below the detected audio voltage of a 30 watt carrier modulated 90% with a 1000 Hz sine wave. This measurement shall be taken with a detector having an effective audio bandwidth of 6 kHz.

3.5.1.7.6.1.6 Incidental FM - The peak deviation arising from undesired frequency modulation shall not exceed 2000 Hz with amplitude modulation of 90 percent at 1000 Hz.

3.5.1.7.6.1.7 Clipping Level - Provisions will be made to allow the modulation clipping level to be adjusted between 70% and 95% with audio input as specified in paragraph 3.5.1.7.6.1.1.

3.5.1.7.6.1.8 Sidetone - The transmitter shall develop a sidetone audio signal which shall appear at the receiver output terminals. The maximum available sidetone output shall be adjustable between 50 milliwatts and 200 milliwatts under standard operating conditions.

3.5.1.7.6.2 FM Modulation (Link 11) -

3.5.1.7.6.2.1 Level - The FM modulator shall be capable of (and adjusted for) a peak deviation of ±30 kHz with an input signal level of 3 V RMS at 1.93 kHz.

3.5.1.7.6.2.2 Response -

3.5.1.7.6.2.2.1 Frequency - From 300 Hz to 3000 Hz, the input signal required to produce 6 kHz deviation shall not depart more than 2 dB from a minus 6 dB per octave straight line (on a dB amplitude vs. log frequency plot), passing through a reference at 1000 Hz. Test equipment linearity error shall not be included.

3.5.1.7.6.2.2.2 Stability - The instantaneous frequency error shall not exceed 0.5 Hz or 2.38 degrees phase shift in a 22 millisecond period at the demodulated transmitter output.

3.5.1.7.6.2.3 Distortion - Any intermodulation product resulting from a two tone test at 10 kHz peak deviation per tone shall be at least 31 dB below either test tone.

3.5.1.7.6.2.4 Input Impedance - The input impedance shall be 600 ohms ±10% balanced, ungrounded with a center tap.

3.5.1.7.6.2.5 Noise Level - The audio voltage detected from an unmodulated 100 watt carrier shall be at least 45 dB below the detected audio voltage of a 100 watt carrier deviated 30 kHz peak with a 1.93 kHz sine wave. This measurement shall be taken with a detector having an effective audio bandwidth of 55 Hz at any audio frequency from 600 to 2200 Hz, and at 2915 Hz.

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3.5.1.7.6.2.6 Incidental Amplitude Modulation - Incidental amplitude modulation shall not exceed 5% when the deviation is ±30 kHz peak.

3.5.1.7.6.3 FSK Modulation (Link 4) -

3.5.1.7.6.3.1 Level - The modulation capability shall be such that +5.0 volts applied to the data input terminals shall increase the carrier frequency 20 kHz (±10%) and -5.0 volts applied to the data input terminals shall decrease the carrier frequency 20 kHz (±10%).

3.5.1.7.6.3.2 Response - The peak-to-peak frequency deviation shall not vary more than ±2 dB with a sine wave modulated signal applied to the data input terminals of constant level over the frequency range from 300 to 5000 Hz. Peak deviation shall be adjusted to 20 kHz at 1000 Hz.

3.5.1.7.6.3.3 Distortion - The FSK modulated rf frequency shall be linear within ±5% of the maximum peak-to-peak frequency deviation over a range of data input voltage levels between 0 and ±5.0 volts.

3.5.1.7.6.3.4 Input Impedance - The data input circuit of the transmitter shall present a grounded resistive load of 2200 ohms ±10% to the signal source.

3.5.1.7.6.3.5 Noise Level - Without modulation, the frequency deviation developed by internal noise shall not exceed 5% of the maximum peak-to-peak deviation of the data signal.

3.5.1.7.6.3.6 Incidental Amplitude Modulation - Incidental amplitude modulation shall not exceed 5% when the deviation is ±20 kHz.

3.5.1.7.6.4 Sonobuoy Command Modulation - This is a single fixed frequency derived electronically from the frequency standard of paragraph 3.4.2.2. The solid-state transmitter shall be capable of producing into a 50 ohm resistive load, 12.6 watts of carrier power -1, +3 dB during Sono transmission. Under service conditions, the carrier power output shall not vary more than ±3 dB from the nominal value.

3.5.1.7.6.4.1 Input Level (Sono) - The rf carrier frequency shall be amplitude modulated not less than 80% nor more than 100% when a single sinusoidal signal having a frequency from 1000Hz to 50 kHz and having an amplitude of 0.5 volts RMS is applied at the Sono input terminal. When multiple tones are

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provided for transmission, the instantaneous peak value of the sum of all tones applied at the Sono input terminal shall not exceed 0.7 volts.

3.5.1.7.6.4.2 Phase Stability - The phase stability of the transmitter shall be within 1.5 degrees at any one frequency over the 35 to 65 kHz band.

3.5.1.7.6.4.3 Distortion -

a. Intermodulation - Intermodulation distortion products resulting from the combination of two equal amplitude discrete frequency (non-harmonically related) tones shall be measured when the rf output level is 12.6 watts -1, +4 dB, and shall be at least 30 dB below the amplitude of the desired discrete frequency outputs.

b. Harmonic - The total harmonic distortion of the detected rf output envelope shall not exceed 10% for any tone in the 1000 to 50,000 Hz range when the equipment is adjusted to 80% modulation at 1000 Hz and an output level of 12.6 watts carrier -1, +3 dB.

3.5.1.7.6.4.4 Input Impedance - The input impedance presented by this transmitter to the Sonobuoy Command Generator signal shall be a 100 ohm ±10% resistive, grounded load.

3.5.1.7.6.4.5 Noise Level (Sono) - The audio voltage detected from an unmodulated Sono carrier shall be at least 40 dB below the detected audio voltage of a 12.6 watt -1, +3 dB carrier modulated 90% with a 1000 Hz sine wave. This measurement shall be taken with a detector having an effective audio bandwidth of 6 kHz.

3.5.1.8 Receiver-Transmitter Controls - The following controls shall be located on the front panel of the receiver-transmitter (see Figure 2).

a. SQUELCH MAIN (Screwdriver Adjustable Potentiometer) - Allows periodic maintenance adjustment of squelch threshold level in Main Receiver.

b. SQUELCH GUARD - Same as a., but for Guard Receiver.

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c. RCVR/XMTR TEST - (Three position, momentary, spring-return to center, toggle switch) - Initiates the self-test sequence when held in either RCVR TEST or XMTR TEST position.

d. SELF TEST - (Seven position rotary switch) - Used in conjunction with RCVR/XMTR TEST toggle switch to initiate self-test of the overall receiver-transmitter or each functional assembly. The seven test positions shall be as follows:

(1) OVERALL R/T
(2) POWER SUPPLY
(3) FREQUENCY SYNTHESIZER
(4) RECEIVER-MAIN
(5) RECEIVER-GUARD
(6) TRANSMITTER
(7) POWER SUPPLY OVERLOADED

3.5.1.9 Indicators - The following indicators shall be located on the front panel of receiver-transmitter.

a. Elapsed Time Indicator - Time totalizing meter which shall indicate the total time that power is applied to Radio Set.

b. POWER ON - Illuminates to indicate that power has been applied to Radio Set AN/ARC-143B.

c. TEST GO - Illuminates to indicate a satisfactory condition during a self-test sequence.

3.5.1.10 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type or an Approved Equivalent	Function
J1	Bendix 21-203222-207	Signal
J2	Bendix 21-203216-26P	Power
J3	Type N Jack	Antenna Signal
J4	MS3122E-12-10S	Mike/Headset Receptacle
J5	MS3122E-12-10P	Blower Control/ Power
J6	RCA 8781905-1	70 MHz Input
J6	RCA 8781905-1	70 MHz Output
J6	RCA 8781905-1	LO Output
J6	RCA 8781905-1	1 MHz Ref. Input

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3.5.2 Radio Set Control C-7791B/ARC-143 - The radio set control (WRA) shall operate with an auxiliary control for satellite mode pre-empting and shall meet the following requirements:

3.5.2.1 Function - The radio set control shall provide all necessary operator controls. It shall be possible to perform the following functions:

a. Switch the radio set on and off, by a hard-wired switch closure.

b. Mode Selection - T/R (AM Voice), T/R + G (AM Voice plus Guard Receiver), FM, FSK, ADF (Automatic Direction Finding), Sonobuoy Transmission, Guard, Test R (Overall Receive), Test T (Overall Transmit). Additionally, the radio set control shall provide an indication of overall receiver-transmitter GO/NO GO status when in the TEST R or TEST T modes.

c. Frequency selection of any frequency within the range of radio set.

d. Means to disable the squelch of the Main and/or Guard receiver.

e. When the SAT mode is selected using an auxiliary control,the C-7791 Radio Set Control shall provide ON-OFF switching of the Radio Set as in paragraph a., and selection of the transmitter frequency. The internal Main and Guard receivers and transmitter modulation circuits shall be disabled and the 70 MHz transmitter and receiver 50 ohm interfaces with the external satellite mode shall be enabled by the SAT mode override.

3.5.2.2 Form Factor - The radio set control shall be enclosed in a volume 6.0 inches deep, 5.75 inches wide, and 3.0 inches high, including the front panel extensions and the rear connector. (See Figure 3.)

3.5.2.3 Weight - The weight of the radio set control shall not exceed 2.3 pounds.

3.5.2.4 Contents - The radio set control shall contain circuits necessary to provide manual frequency selection, mode control,squelch disabling of Main and/or Guard receivers plus the indication of overall GO/NO-GO system testing and status indication of the radio set. The above control information shall be

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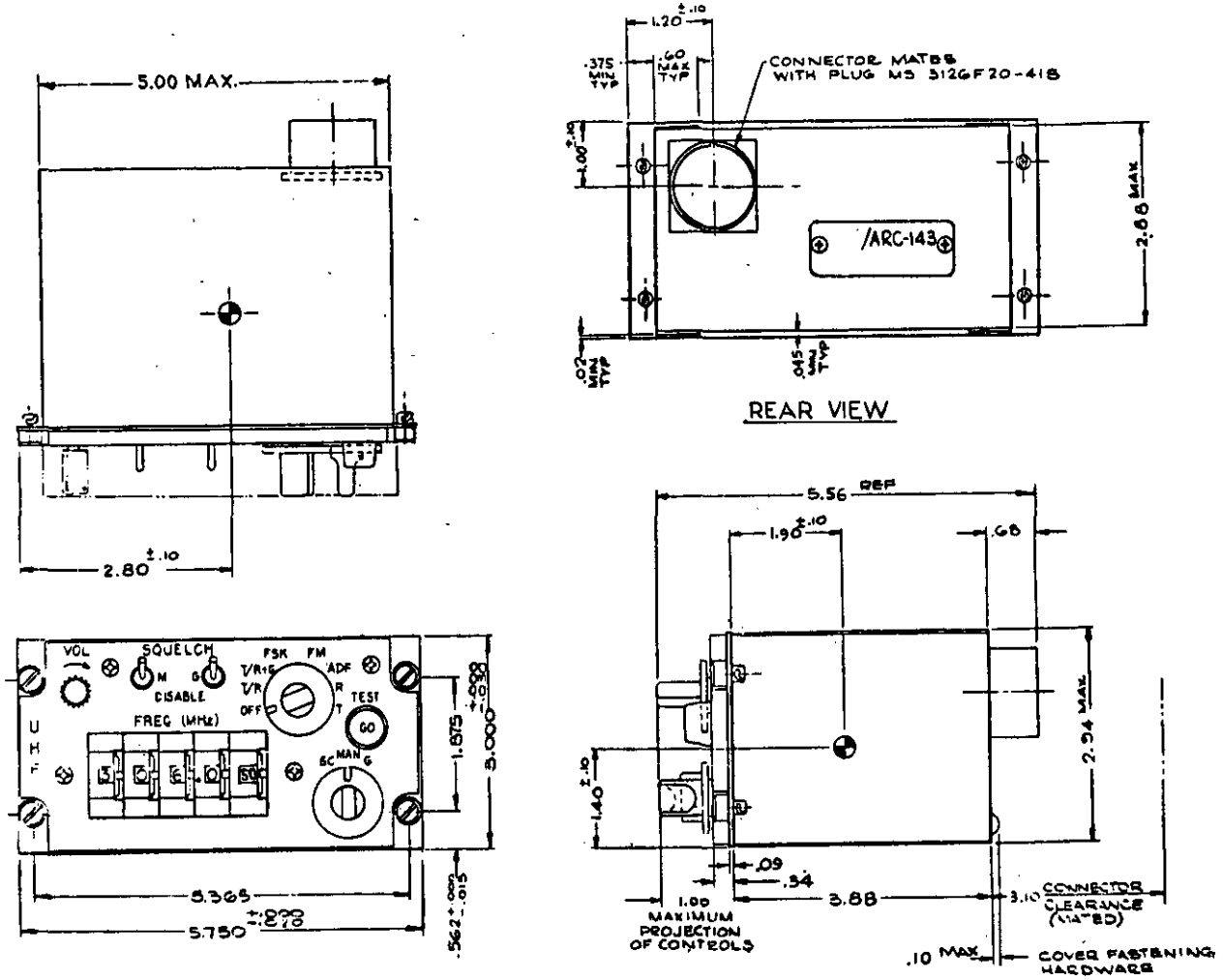


Figure 3. Outline and Mounting for Radio Set Control, C-7791B/ARC-143

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formatted into a 32-bit message and transferred between the radio set control and the transceiver in a serial bit stream via a pair of control wires. The necessary message timing shall also be generated within the control, radio set and sent to the receiver-transmitter via an additional wire.

3.5.2.5 Computer Control - The radio set shall be capable of being controlled by a general purpose digital computer provided the message structure, timing and synchronization is the same as that generated by the radio set control. A single wire control line having a ground (+0.2) volts shall release the control unit and permit computer takeover of the radio set. An open circuit on this control line enables the radio set control to operate the radio set.

3.5.2.5.1 Mode Modify and FM Select Functions - Two lines shall be accepted by the radio set control, "Mode Modify" and "FM Select", and shall force the radio set to operate as follows:

a. MODE MODIFY - This line will present a ground (OFF) or an open circuit (ON) to the radio set control.

b. FM SELECT - This line shall provide a signal of 0 (+0.4, -0) volts capable of sinking 10 ma (OFF) or 5 (+0.5, -1.4) volts DC (ON).

Operation - When the "Mode Modify" line is grounded, the "FM Select" line shall have no effect and receiver-transmitter mode of operation shall be determined by switch positions on the radio set control front panel.

When the "Mode Modify" line is open circuit (ON), the receiver-transmitter mode shall be determined by combination of radio set control front panel switch selection and signal present on the "FM Select" line. Receiver-transmitter modes shall be controlled as follows:

(1) Selection of Sono Command Mode, Guard Mode, Test R or Test T on the radio set control front panel switches shall force the receiver-transmitter mode to the radio set control selected mode, independent of the signal present on the "FM Switch" Line.

(2) Selection of T/R, T/R+G, ADF, FSK or FM on the radio set control front panel while the SC/M/G switch is in M and 5 (+0.5, -1.4) volts DC is on the "FM Select" line shall force the receiver-transmitter to operate in the FM mode.

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(3) Selection of T/R, FSK or FM on the radio set control front panel while the SC/M/G switch is in M and 0 (+0.4, -0) volts DC is on the "FM Select" line shall force the receiver-transmitter to operate in the T/R mode.

(4) Selection of T/R+G or ADF on the radio set control front panel while the SC/M/G switch is in M and 0 (+0.4, -0) volts DC is on the "FM Select" line shall force the receiver-transmitter to operate in the T/R+G mode.

3.5.2.6 Panel/Mounting - Control, Radio Set shall be of the console mounting type in accordance with MIL-C-6781 for Type 1 controls. The integrally illuminated panel shall be in accordance with MIL-P-7788.

3.5.2.7 Controls - The following controls shall be provided and shall be located on the front panel for operator adjustments:

a. Frequency Selectors - Enable selection of transceiver operating frequency from 225.000 MHz thru 399.975 MHz in .025 MHz steps (7000 possible channels).

b. VOL (Potentiometer) - Controls audio output levels of Main and Guard receivers.

c. SC-MAN-G Switch - Operates in conjunction with the Mode Selector switch to determine operating condition: SC (Sono Command), MAN (Manual), or G (Guard).

d. Mode Selector Switch (Eight Position Rotary) - Establishes one of the following modes of operation: OFF, T/R, T/R+G, FSK, FM, ADF, TEST R or TEST T.

e. Main Squelch Disable (Toggle Switch) - Permits disabling the Main receiver squelch.

f. Guard Squelch Disable (Toggle Switch) - Same as e., except for Guard receiver.

3.5.2.8 Indicators - The following indications shall be provided and shall be located on the front panel for operator interpretation:

a. FREQ (MHz) Display - Operates in conjunction with the frequency selectors to indicate the operating frequency of the radio set.

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b. GO Indicator - Illuminates to indicate that the overall receiver or the overall transmitter has passed self-test.

3.5.2.9 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designations	Receptacle Type	Function
J1	MS3112-E-20-41P	Signal/Power

3.5.2.10 Primary Input Power Requirements - The control shall meet the applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits specified therein. The power requirements shall not exceed the specified amount.

a. AC, 5 volts single phase, 10 VA (panel lighting)

b. DC Power, 18 volts, 1 watt (Panel GO Lamp)

The control unit shall not malfunction under the emergency voltage limits specified in MIL-STD-704.

3.5.2.11 Cooling Air Requirements - None shall be required.

3.5.3 Mount, Receiver-Transmitter MT-4064/ARC-143 - The Mount Electrical Equipment MT-4064 (WRA) shall meet the following requirements:

3.5.3.1 Function - The Mount, Electrical Equipment shall provide isolated mounting for Receiver-Transmitter, Radio, RT-932B/ARC-143.

3.5.3.2 Form Factor - The Mount, Electrical Equipment envelope shall not exceed 21.6 inches long (excluding clamping screws) by 6.5 inches wide and 2.8 inches high, including isolators. An allowance of an additional 1.75 inches maximum on all sides shall be provided for sway space except a minimum of 10 inches shall be provided in length to accommodate the cooling blower and air intake which is physically attached to the receiver-transmitter. (See Figure 4.)

3.5.3.3 Weight - The weight of the MT-4064/ARC-143 shall not exceed 3.5 pounds.

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3.5.3.4 Contents - Mount, Electrical Equipment shall contain vibration and shock isolation means, applicable holddowns and mounting guides for the receiver-transmitter. The mount shall contain flexible straps for electrical bonding in accordance with MIL-B-5087.

3.5.4 Amplifier, Radio Frequency AM-6699/A - The SAT preamplifier (WRA) assembly (not part of Radio Set AN/ARC-143B) shall operate with the radio set where a single satellite antenna is used.

3.5.4.1 Function - In the SAT mode, the satellite preamplifier assembly shall route the ARC-143B transmitter rf output to the satellite antenna. The signal received from the satellite shall be routed through the bandpass filter and low noise preamplifier prior to input into the ARC-143B R/T unit.

3.5.4.2 Form Factor - The SAT preamplifier shall be enclosed in a case with overall dimensions not exceeding 2 x 9.5 x 10.5 inches (see Figure 5).

3.5.4.3 Weight - The weight of the preamplifier shall not exceed 3 pounds.

3.5.4.4 Contents - The SAT preamplifier assembly shall contain the following subassemblies:

a. Bypass Switch - The bypass switch shall control the routing of signals inside the SAT preamplifier assembly. In the SAT transmit mode, the transmitter signal entering the SAT preamplifier assembly shall be routed directly to the SAT antenna. In the SAT receive mode, the received signal at the SAT antenna shall be routed into the bandpass filter and low noise amplifier. The output of the low noise amplifier shall be routed to the output port of the assembly. The mutually exclusive selection of either routing accomplished by the bypass switch shall be controlled by a receive-transmit switch bias from the receiver-transmitter unit.

b. Bandpass Filter - The bandpass filter shall accept the received satellite signal routed from the satellite antenna via the bypass switch. The output of the bandpass filter is connected to the low noise amplifier.

c. Low Noise Amplifier - The low noise amplifier accepts the received satellite signal from the bandpass filter, amplifies it and outputs it to the bypass switch which shall route it to the LOS/SAT antenna switch.

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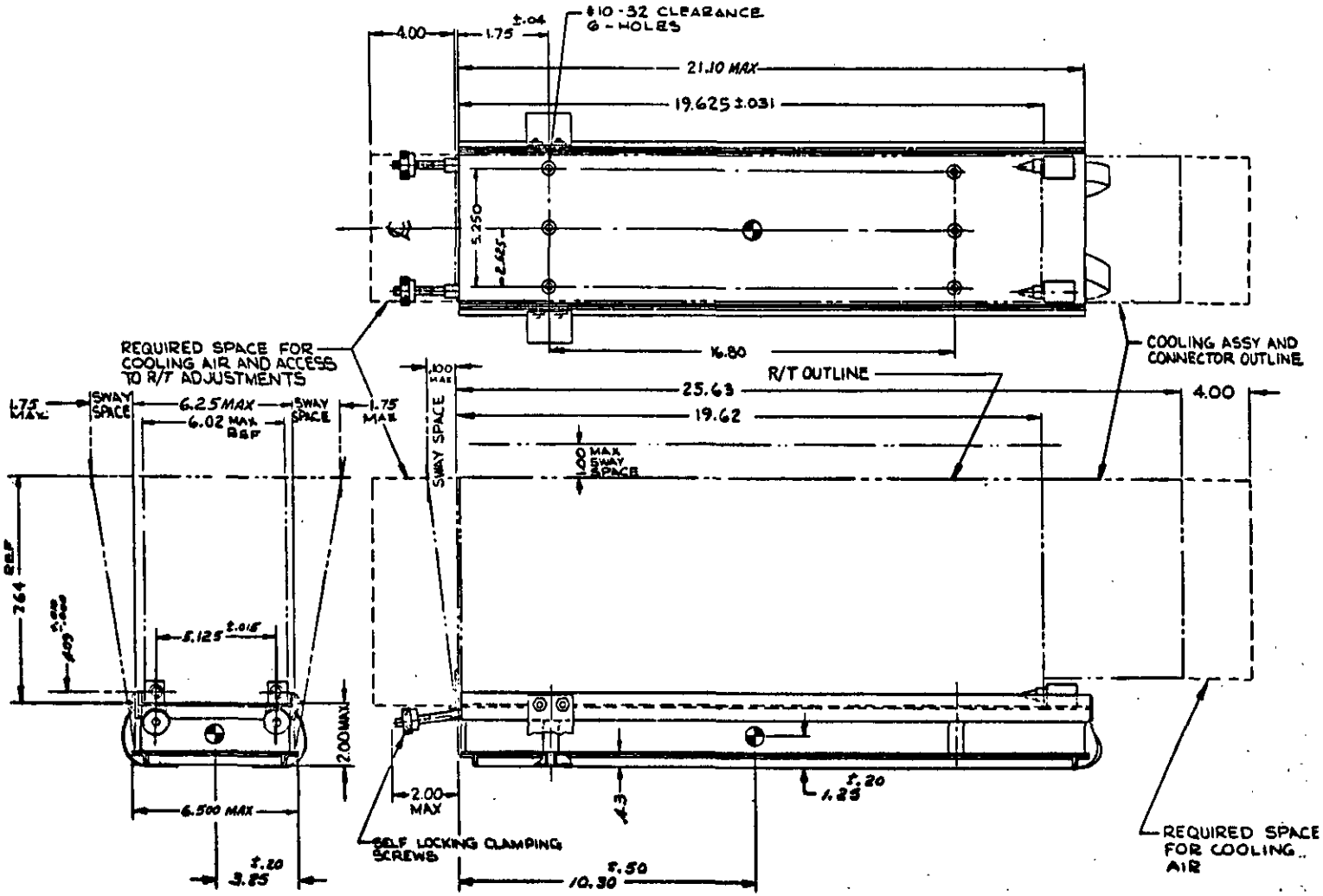


Figure 4. Outline and Mounting for Mount, Electrical Equipment MT-4064/ARC-143

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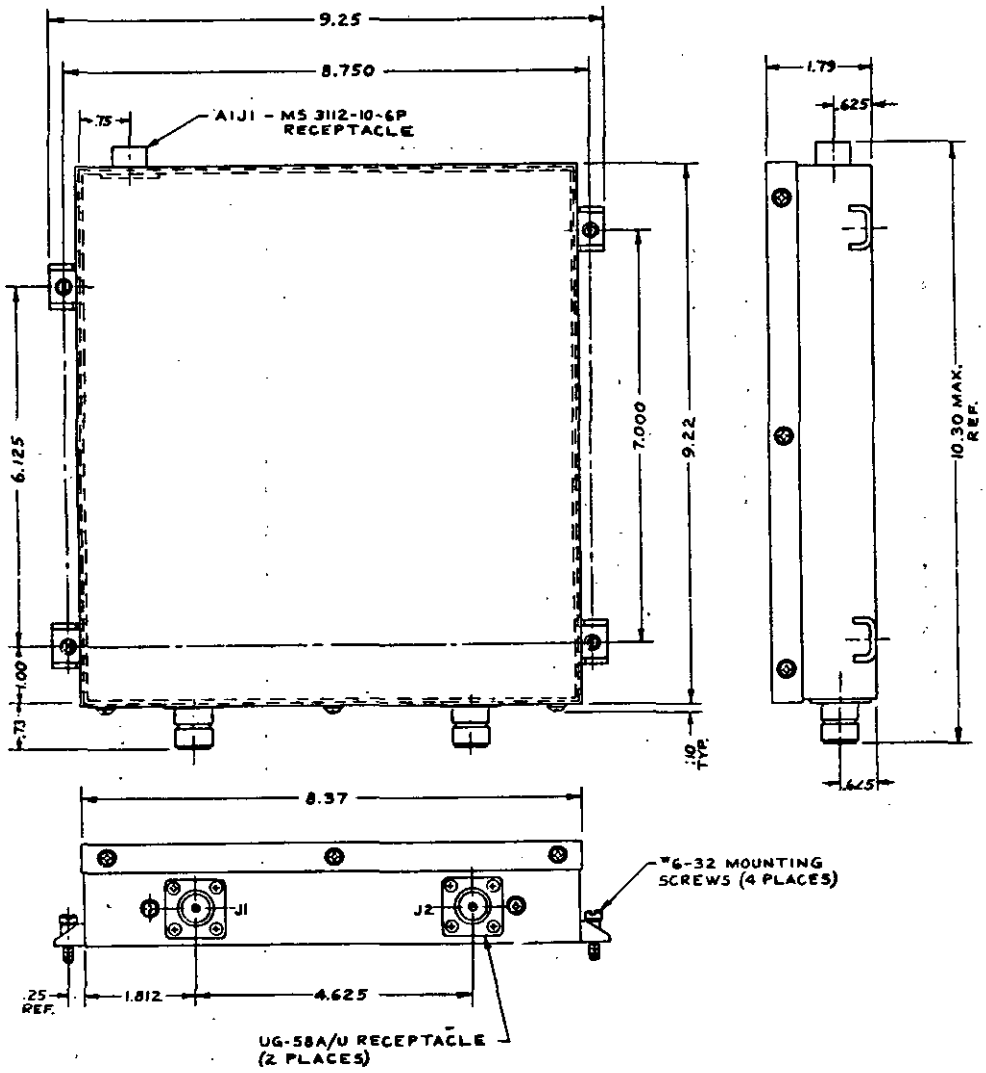


Figure 5. Outline and Mounting for Amplifier, Radio Frequency, AM-6699/A

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3.5.4.5 Characteristics -

3.5.4.5.1 Noise Figure - The noise figure of the SAT pre-amplifier assembly in the receive mode shall not exceed 4.5 dB under standard conditions and shall not exceed 5.5 dB under service conditions.

3.5.4.5.2 Bandpass Characteristics - The receive response measured at the output port of the assembly at 250 MHz and 270 MHz shall not be more than 1 dB below the nominal center frequency response and the peak-to-peak ripple shall not exceed 1 dB within the passband.

3.5.4.5.3 Gain - The minimum gain within the bandpass (as defined in Paragraph 3.5.4.5.2)shall be 22 dB.

3.5.4.5.4 Terminations - In the satellite receive mode the input impedance as measured at the satellite antenna connector shall be 50 ohms with a VSWR of no more than 2.5 to 1 at any frequency between 250 MHz to 270 MHz when the output load at the connector for the LOS/SAT switch is 50 ohms with a VSWR of less than 1.05.

3.5.4.5.5 Electrical Connections -

Reference Designations	Receptacle Type	Function
J1	Coaxial Jack Type N	RF Satellite Antenna
J2	Coaxial Jack Type N	RF Amplifier Output
J3	MS3112E10-6P	Control/Power

3.5.5 SAT/LOS Select Panel Assembly - The satellite/line-of-sight select panel assembly (not part of Radio Set, AN/ARC-143B) shall operate with the radio set in the ground based Tactical Support Center, and its exact configuration may vary.

3.5.5.1 Function - The SAT/LOS select panel assembly shall act as the auxiliary control in the satellite mode override operation (see 3.5.2.1). In the LOS position, it shall permit normal operation of the LOS functions of the radio set. In any of the 3 SAT positions, it shall enable operation with the external modem via the 70 MHz ports and provide for selection of any of the three satellite frequency offsets (see 3.4.1 and 3.4.7.8).

3.5.5.2 Form Factor - The panel shall be of 0.125 thick aluminum sheet and shall not exceed 5.25 inches by 19.015 inches by 3 inches (see Figure 6).

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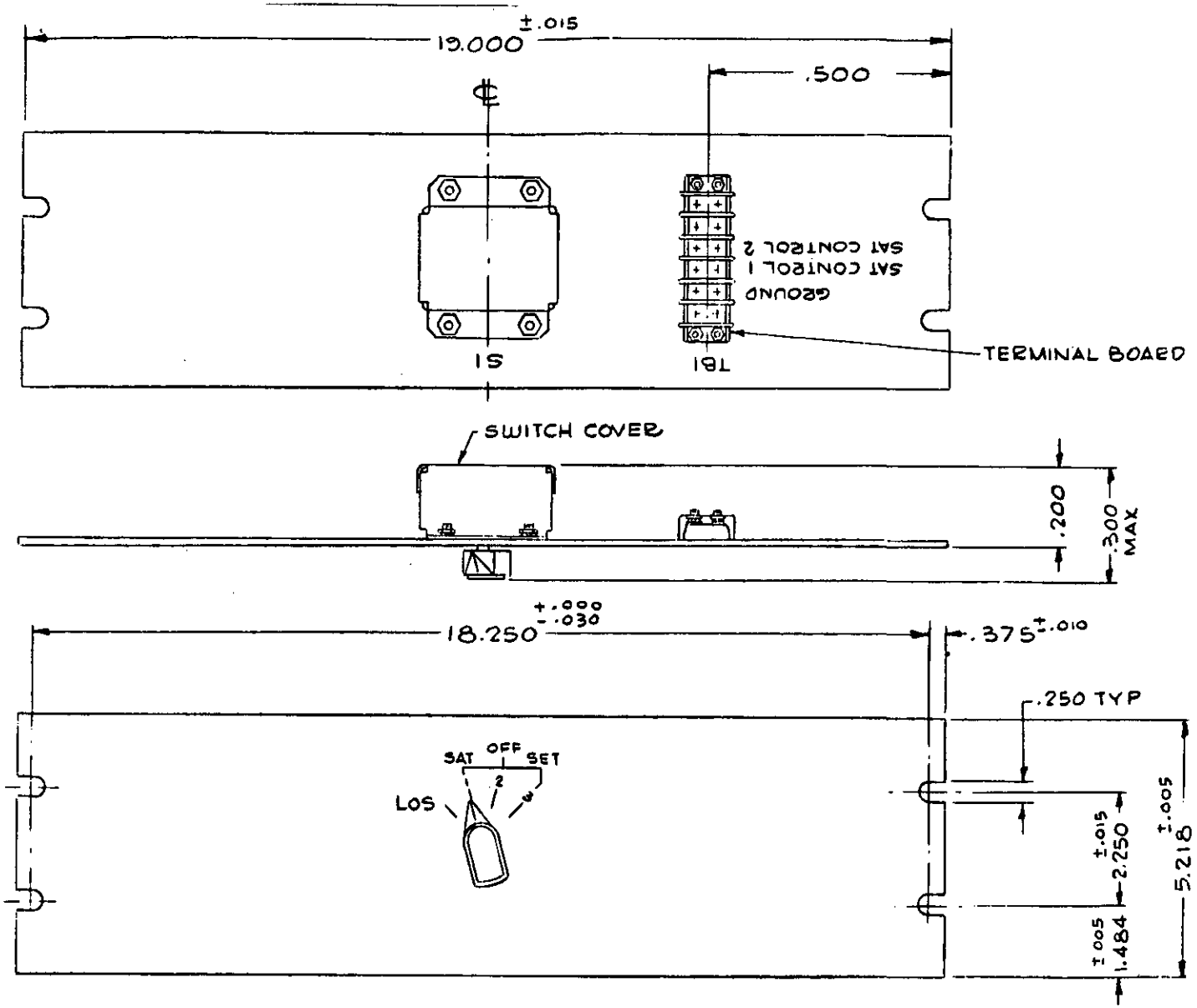


Figure 6. Outline and Mounting for SAT/LOS Select Panel

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3.5.5.3 Weight - The weight of the SAT/LOS select panel assembly shall be a maximum of 2 pounds.

3.5.5.4 Characteristics -

a. Switch - A single, detented, 4 position rotary switch shall be provided for the selection of modes and when read in a clockwise direction, the positions shall be marked as follows:

LOS, SAT OFFSET 1, 2, 3

The purpose of the rotary switch is to provide circuit closures between wire 3 and wire 1 or wire 2 or both wires, in accordance with 3.4.7.8.

b. Terminal Board - A terminal board (TB1) shall be provided on the back side of the panel to permit three wire connections to the rotary switch as follows:

WIRE 1	SAT CONTROL 1
WIRE 2	SAT CONTROL 2
WIRE 3	GROUND

3.6 Built-In Test - The radio set shall have a self-contained built-in test which permits overall dynamic system testing in less than 3 seconds.

3.6.1 Overall System Test - Overall system testing is initiated from the control, testing the receiver circuitry first and then the transmitter circuitry. A "GO"-"NO-GO" indication is displayed to the operator on the Control.

3.6.2 Fault Isolation from the Receiver-Transmitter - The "GO"-"NO-GO" indicator and a switch permits the operator to duplicate the overall R/T test as initiated from the control. Thus, malfunction of the control unit itself can be determined.

The "GO"-"NO-GO" indicator and a 7 position switch (with a self-test legend) provides the means of determining which of the functional assemblies of the receiver-transmitter is malfunctioning).

By sequentially switching, fault isolation to the following is possible:

a. Power Supply
b. Frequency Synthesizer
c. Receiver, Main
d. Receiver, Guard
e. Transmitter

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Given a singular faulty circuit in the R/T, the use of proper procedures and the self-test circuitry will correctly isolate a fault to the faulty functional assembly with a minimum probability of 80%.

All required "on aircraft" tests shall be accomplished with built-in test and/or ancillary aerospace ground equipment (AGE). "On aircraft" tests permitted include performance verification of R/T and control, fault isolation to the malfunctioning R/T or control, and use of built-in test.

3.6.3 Receive and Transmit Monitoring - The following operational capabilities are available through the Headset/Microphone connector referred to in 3.4.7.9.

a. Key the transmitter
b. Voice modulate the transmitter (AM)
c. Monitor the receiver output (AM)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to those tests spelled out in the contract to determine compliance with requirements. Appropriate tests will be selected from the following:

Test	Paragraph
a. Preproduction (First Article) Tests	4.2
b. Initial Production Tests	4.3
c. Production Acceptance Tests	4.4
d. Life Test	4.5

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under

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the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a Government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, constructions, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the pre-production sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on the equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

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4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the Initial Production Test the equipment shall be reworked by the contractor by replacing all wear on damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the courses of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests
b. Sampling Tests
c. Reliability Assurance Tests
d. Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product
b. Operational Test
c. Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of ten minutes prior to the beginning of the ten hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the equipment in two directions, the vertical direction shall be used. During the ten hour period of operation following the ten minute vibration period, the equipment shall be

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mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The ten hour period specified above may be composed of two five hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipment shall be selected for sampling tests by the Government inspector in accordance with the following:

Quantity of Equipment Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	0 #
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Sampling Tests are not required when Reliability Assurance Tests are conducted.

NOTE: If by contract action, the Initial Production Test is deleted, a Sample Test shall also be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

c. Manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However,

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each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Test - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For Qualification Phase, Test Level E shall be used. The Accept-Reject Criteria for Test Plan III shall be used.

4.4.3.2 Reliability Production Acceptance (Sampling) Phase Tests - The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". Test Level E shall be used.

4.4.3.2.1 All Equipment Test - Each equipment produced except those submitted for the Reliability Qualification Test, shall be tested for 50 hours. Prior to the 50-hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures.

To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totalled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

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4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

d. Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life tests shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Condition - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal Room
Altitude	Normal Ground (0 - 5000 ft.)
Humidity	Room Ambient
AC Voltage	115 +5 volts (at lowest applicable frequency)
DC Voltage	28 VDC
DC Voltage	5 VDC (Panel Lighting)

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

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4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

a. Part type number

b. The circuit reference symbol number

c. The part function

d. Name of the manufacturer

e. Nature of the failure

f. The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

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4.8 Resubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - See Paragraph 3.4.1.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

a. Title, number, and data of this specification.

b. Selection of applicable levels of packaging and packing (see 5.1).

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any specification.

b. This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specification where applicable, shall be specifically approved in writing by the procuring activity.

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c. Reference Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Non-Repairable Subassemblies - As a general rule, non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-To-Failure greater than 5,000 hours and for many specifications this figure must be near 50,000 hours.

6.6 Type Designations - The type designation may be changed by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Revisions - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

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6.8 Associated Equipment - The Radio Set shall operate with the following associated equipment:

a. AN/AIC-22 or equivalent

b. Data Terminal Set AN/ACQ-5

c. Direction Finding Group AN/ARA-25, AN/ARA-48, or AN/ARA-50

d. Sono Command Generator (CASS)

e. TTY Signal Data Converter

f. Voice Security Unit (KY-28)

g. CALAC Communicator Selector (A345)

h. CALAC Communication Switching Matrix (A346)

i. Antenna Switch (740645-101)

j. Satellite DPSK Modem

k. Memory Storage Buffer

l. Integrated SATCOM Control

m. Satellite Power Amplifier

n. Integrated Satellite Antenna Network

6.9 This specification is under the cognizance of AIR-53341E.

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-81628B(AS) RADIO SET AN/ARC-143B

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. *GIVE PARAGRAPH NUMBER AND WORDING.*

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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